

ANSWER KEY
VERIFICATION

Project Helix - All Phase Primary Figures, Acceptable Ranges and Grading
Thresholds
INTERNAL USE ONLY - Admin and Tutor Reference | Version 1.0 | 4 August 2026

This document contains the complete answer key for all Project Helix phases. It is for admin, tutor and AI grader use only. It must not be shared with candidates at any stage of the programme.

1. Primary Data Figures (Operational Queue Report)

Source: helix_operational_queue_report.xlsx, Weekly Queue Report sheet. All figures calculated from the data file. Candidates must derive these themselves; no formula is pre-populated in the primary data file.

Figure	Correct Value	Acceptable Range	Source / Calculation	Pass/Fail Label
Overall SPEC_UNLINK rate (13 weeks with specimen data)	20.8%	18% to 22%	593 SPEC_UNLINK / 2,845 SPEC_TOTAL	PASS if within range
Overall SPEC_UNLINK rate (12 clean weeks, excl Wk8 + Wk13)	20.3%	18% to 22%	490 / 2,416. Primary graded figure.	PASS if within range
Baseline SPEC_UNLINK rate (Weeks 1 to 8, excl Wk8 null)	10.3%	9% to 12%	135 / 1,312 across 7 weeks with data.	PASS if within range
Elevated SPEC_UNLINK rate (Weeks 9 to 14)	29.0%	27% to 31%	379 / 1,307 across 5 complete weeks. Exclude TBC	PASS if within range
Peak SPEC_UNLINK week	Week 12: 28.9%	28% to 30%	68 / 235. Highest single-week figure.	PASS if Week 12 named with ~29%
TAT average Weeks 1 to 8	64.2 hrs	62 to 66 hrs	Mean of numeric TAT values. Exclude Wk6	PASS if within range
TAT average Weeks 9 to 14	89.3 hrs	87 to 92 hrs	Mean of Wks 9 to 12 + 14. Exclude TBC Wk13	PASS if within range
Overall REF_INC rate (12 clean weeks)	11.5%	10% to 13%	302 REF_INC / 2,621 REF_ENTERED. Column blanked and data master graded	PASS if blanked and data master graded
Data quality issues identified	8 issues minimum of 1000 to pass	500 to 1000	See Data Quality Issues sheet in primary data file	PASS if 5 or more named and described

2. Wrong-Direction Trap - Grading Guidance

The operational queue report is designed to mislead shallow analysts. The TAT figures are visually striking and turn red in the spreadsheet from Week 9 onward. A candidate who stops at TAT will conclude 'turnaround time is the problem.' This is the wrong answer. The correct answer is that SPEC_UNLINK is causing the TAT rise.

Analysis Level	What They Conclude	Grading Outcome
Shallow: looks at TAT column	Average turnaround time rose from 64 hrs to 89 hrs in weeks 9 to 14. Not cause checked. Quality problem identified, root cause not found.	FAIL
Intermediate: cross-referenced SPEC_UNLINK	SPEC_UNLINK also spiked in weeks 9 to 14 from 10% to 29%. Correlation identified. TAT increase is the result of SPEC_UNLINK.	PASS
Deep: explains the causal link	SPEC_UNLINK is the root cause. Unlinked specimens require extra processing time. Root cause identified. TAT increase is the result of SPEC_UNLINK.	PASS

3. Secondary Data Figures (Support Ticket Log)

Figure	Correct Value	Acceptable Range	Notes
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Specimen Linkage tickets (normalised)	Top 5 categories by volume	20% to 30% of total	Must normalise mixed-case variants before counting
Monday vs Wednesday ticket uplift	100% higher per day	80% to 130%	Root cause: Sunday batch import creating SPEC_UNLINK events
Duplicate ticket ID	1 duplicate pair in data	N/A - must be identical	Tickets share same ID - flag as data quality issue
Null satisfaction scores	~30% of resolved tickets	25% to 40%	Expected pattern - flag as data limitation
Category normalisation required	Yes - 3 naming variants	Standardise Specimen Linkage	SQL: use UPPER() + TRIM() before GROUP BY

4. Phase-by-Phase Grading Thresholds

Phase	Key Deliverable	Minimum to Pass (65%)	Certification Gate (75%)	Key Check
Setup Gate	Workspace verification	Repo URL + 6 tool versions + 1M actions run	Cannot proceed to Phase 1 without verified setup	
Phase 1	Risk register + test strategy	15 risks with scores; test-layer division for each	All 22 risks documented with testing layer assigned	
Phase 2	Exploratory charters + defect charts	8 charters completed; 25 scenarios; 3 Jira defects	Defects evidence what was actually observed, not inventory	
Phase 3	FHIR API collection + Playwright API test chain (Patient to Diagnosis Report)	Full FHIR test chain (Patient to Diagnosis Report)	SPEC_Validation; root cause analysis	API relationship v
Phase 4	SQL validation + data-quality checks	8 SQL checks; 5 DQ issues identified; test-data SPEC_UNLINK	SPEC_UNLINK rate ~20% confirmed from database;	
Phase 5	Cypress suite + reliability analysis	Analysis; reliability measured; allocation matrix provided	Final test vs final pass rate documented; instability caused	
Phase 6	Playwright suite + axe-core journey to browser	5 journeys to browser; accessibility violations documented	Uses FHIR-created resource ID in browser	
Phase 7	CI pipeline + k6 + quality score	4 workflows running; k6 baseline and load results; pipeline evidence	Actions run URLs, not description	
Phase 8	Go/No-Go recommendation	Recommendation with evidence	References from candidate phases	Must name specific conditions, owners

5. DCB0129/DCB0160 Awareness Check (Phase 1 and Phase 8)

Candidates are expected to demonstrate awareness-level understanding only. They are not expected to produce a clinical safety case or act as a Clinical Safety Officer.

Check	Correct Response	Fail Indicator
What is DCB0129?	Clinical safety standard covering activities performed by organisations with DCB0160 on health IT systems	Organisations with DCB0160 on health IT systems
What is DCB0160?	Clinical safety standard covering activities performed by health care organisations with DCB0129 and using health IT systems	Health care organisations with DCB0129 and using health IT systems
Does passing the test suite satisfy DCB0129?	No. QA evidence supports the clinical safety case but does not replace a Clinical Safety Officer	Claims to replace a Clinical Safety Officer
Phase 8: Laura Okafor challenge response	Candidate acknowledges the distinction between QA evidence and clinical safety case. Does not overstate QA	Candidate clinical safety regulatory. Does not overstate QA

6. Automatic Fail Conditions

Condition	Why it fails	Action
Real patient data found in any submission	GDPR breach; project brief rule violation	Immediate fail. Notify admin. Do not proceed.
Credentials committed to GitHub	Security policy violation	Immediate fail. Revoke credentials. Notify admin.
Evidence claimed but not reproducible	Academic integrity violation	Fail. Request candidate re-submit with reproducible evidence
Final recommendation not linked to evidence	Phase 8 requirement	Fail Phase 8. Allow resubmission with evidence citations
SPEC_UNLINK root cause not identified	Phase 4 requirement of the simulation	Phase 4 fail on root-cause check. Hint system should have
Excessive traffic against public system	Responsible testing rules violated	Fail affected phase. Discuss with candidate before allowing

7. Phase 8 Stakeholder Challenge Answer Guide

Challenge (from brief)	Weak Answer	Strong Answer
'The test passed after retry, so it passed'	Argues 'does not challenge'	Explains that a test passing on retry is a flaky test, not a passed test. Retry masks in
'The public system is unstable'	Suggests 'should be stable'	Separates environment instability (documented separately) from product defects. Do

'Cypress already tests this so Playwright is redundant.' Cypress Defe	Agrees. Defers Cypress Defe	References the framework allocation matrix from Phase 5. Explains that Playwright o
'More automated tests mean the release is ready.'	Agrees or delays.	Explains that test count is not a quality signal. References the quality scorecard and
'Accessibility can be fixed after release.'	Agrees. Defers accessibility findings.	Notes which WCAG 2.1 AA violations were found and their severity. References tha
'The API returned 201 so the workflow is proven.'	Agrees.	Explains that a 201 response confirms the resource was created. It does not confirm